Work in progress — draft, not peer reviewed

A Lean-Verified Conditional Reduction of Periodic Navier–Stokes Regularity on $\mathbb{T}^3$ to a Single Integrability Hypothesis

Conditional results; draft — public repository, not peer reviewed

Benjamin Stanley Frohman
ORCID 0009-0006-7068-3718 · frohmanbenjamin@gmail.com
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Scope. This paper reports a *conditional* result. It does **not** solve the Clay Millennium Problem, and it does not prove global regularity of the 3D Navier–Stokes equations. The single remaining hypothesis, `PeriodicBKMIntegrability`, is a named Prop that is **not proved**. It is equivalent in difficulty to Clay Statement (B).

How to cite / What is claimed. In Lean 4 with Mathlib, we give a machine-checked reduction of Clay Millennium Statement (B) (periodic Navier–Stokes on $\mathbb{T}^3$) to the single hypothesis `PeriodicBKMIntegrability`. We prove this hypothesis is equivalent to global existence of smooth solutions for every smooth periodic initial field. The BKM criterion and H^3 control are proved unconditionally, using only the standard axioms. `PeriodicBKMIntegrability` remains open. The equivalence of vorticity integrability with continuation of smooth solutions is classical (Beale–Kato–Majda, 1984 [1]); the novelty here is the complete, `sorry`-free formalization. Cite as: B. S. Frohman, reduction paper, DOI 10.5281/zenodo.23842295; main paper, DOI 10.5281/zenodo.23842555.

Abstract

In the Lean 4/Mathlib [4, 12] repository `BenFrohman/periodic-ns-classical-continuation` (commit `f9a26c7`), we prove two results with no hypothesis Props:

- H^3 Beale–Kato–Majda control on $\mathbb{T}^3$, `periodicBKMHsControl_three`;
- the periodic BKM blow-up criterion [1], `periodicBKMCriterion`.

We deduce the exact periodic Clay Statement (B) [5], `Clay_B_exact`, *conditionally* on the single named Prop `PeriodicBKMIntegrability` (`clay_B_exact_of_integrability`). The axioms used are only `propext`, `Classical.choice` and `Quot.sound`.

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Keywords. Navier–Stokes equations; torus; Beale–Kato–Majda criterion; conditional regularity; Lean 4; formal verification; work in progress.

1 Introduction

The Beale–Kato–Majda criterion [1] asserts that a smooth solution of the three-dimensional Navier–Stokes equations can cease to be smooth at a finite time T only if $\int_0^T \|\omega(t)\|_{L^\infty}\,\mathrm{d}t = \infty$. Consequently, Statement (B) of the Clay Millennium Problem [5] on the torus is equivalent to the time-integrability of the vorticity sup-norm for every smooth periodic initial field. In this paper we report a machine-checked version of this reduction in Lean 4: the criterion

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and the required H^3 control are proved with no hypothesis Props, and Statement (B) is derived from the single named hypothesis `PeriodicBKMIntegrability`, which remains open. Section 3 tabulates precisely what is stated and what is proved.

2 The periodic chain on $\mathbb{T}^3$

Throughout, $\nu > 0$, u_0 is smooth, periodic and divergence-free, and (u, p) is the maximal smooth solution of

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \qquad \operatorname{div} u = 0, \qquad u(0) = u_0 \quad \text{on } \mathbb{T}^3, \tag{2.1}$$

with existence time T^*; $\omega = \operatorname{curl} u$. The definitions `Clay_B_exact`, `PeriodicBKMCriterion` and `PeriodicBKMIntegrability` come from the dependency `fefferman-clay-navier-stokes`.

Named hypothesis 2.1 (`PeriodicBKMIntegrability` ν; **open**). For every maximal solution and every T, $t \mapsto \|\omega(t)\|_{L^\infty}$ is integrable on $[0, T]$:

$$\int_0^T \|\omega(t)\|_{L^\infty} \, \mathrm{d}t < \infty. \tag{2.2}$$

Theorem 2.2 (`periodicBKMHsControl_three`†; unconditional). *For every* ν, `PeriodicBKMHsControl` ν 3 *holds. That is,*

$$T^* = T < \infty \quad \textit{and} \quad \int_0^T \|\omega\|_{L^\infty} \, \mathrm{d}t < \infty \quad \Longrightarrow \quad \sup_{t \in [0,T)} \|u(t)\|_{H^3} < \infty.$$

Proof route (all formalized). The H^3 energy identity reduces the claim to a commutator bound and a low-frequency vorticity bound.

- The $s = 3$ commutator bound [6] is

  $$-2 \sum_i \int \left(1 - \frac{\Delta}{4\pi^2}\right)^3 v_i \left((v \cdot \nabla)v\right)_i \, \mathrm{d}x \le C \|\nabla v\|_{L^\infty} \|v\|_{H^3}^2$$

  (`periodicPhysicalCommutatorBound`†, `CommutatorAssembly.lean`).

- The low-frequency bound is

  $$\|V_R \nabla v\|_{L^\infty} \le K(1 + \log R) \|\omega\|_{L^\infty} \qquad (R \ge 2)$$

  (`periodicLowFreqVorticityBound`†, `LowFreqAssembly.lean`). It is proved through the Riesz form of Biot–Savart, a smooth dyadic decomposition with uniformly L^1-bounded block kernels, and a uniform L^1 bound for the de la Vallée Poussin kernel.

Grönwall's inequality with the log splitting then closes the bound.

Theorem 2.3 (`periodicBKMCriterion`†; unconditional). *For every* ν, `PeriodicBKMCriterion` ν *holds. If* $\nu > 0$ *and* $T^* = T < \infty$, *then*

$$\int_0^T \|\omega(t)\|_{L^\infty} \, \mathrm{d}t = \infty.$$

Theorem 2.4 (`clay_B_exact_of_integrability`†; conditional). `PeriodicBKMIntegrability` $\nu \Rightarrow$ `Clay_B_exact` ν.

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Remark 2.5 (Equivalence in difficulty). The converse direction also holds. If Statement (B) gives a global smooth solution, then by uniqueness the maximal solution is global and smooth, so $\|\omega\|_{L^\infty}$ is continuous and integrable on every $[0, T]$. Hence `PeriodicBKMIntegrability` is *equivalent* to the periodic Clay Statement (B); it is not a weaker, merely technical residue. This converse, from Statement (B) itself, is argued informally here. What *is* proved in Lean (`periodicBKMIntegrability_iff_globalLifespan`, `IntegrabilityAttack.lean`) is that `PeriodicBKMIntegrability` is equivalent to `PeriodicGlobalLifespan`, i.e. $T^* = \infty$ for all smooth data. Its proof uses `integrableOn_vorticity_of_lt_tstar`: $\int_0^T \|\omega\|_{L^\infty}\,\mathrm{d}t < \infty$ whenever $T < T^*$. This paper is therefore a verified reduction, not a proof of regularity.

3 Stated vs. proved

Statement	Status
H^3 BKM control, `periodicBKMHsControl_three`	proved unconditionally
Commutator bound ($s = 3$), `periodicPhysicalCommutatorBound`	proved unconditionally
Low-frequency vorticity bound, `periodicLowFreqVorticityBound`	proved unconditionally
Periodic BKM criterion, `periodicBKMCriterion`	proved unconditionally
Clay Statement (B), via `clay_B_exact_of_integrability`	proved conditionally on `PeriodicBKMIntegrability`
`PeriodicBKMIntegrability`	not proved (open; equivalent to Clay (B))
Converse: (B) ⇒ `PeriodicBKMIntegrability`	not proved in Lean (informal argument only)
`periodicBKMIntegrability_iff_globalLifespan`†	proved unconditionally
`integrableOn_vorticity_of_lt_tstar`† (below T^*)	proved unconditionally
Small data, `integrability_of_smallDataGlobal`†	proved from named `PeriodicSmallDataGlobal` (not formalized)
Prodi–Serrin, `integrability_of_prodiSerrin`†	proved from named `PeriodicProdiSerrinCriterion` (not formalized)
$\kappa = 0$ route, `periodicBKMIntegrability_of_vorticityAPriori`†	proved from `PeriodicVorticityAPrioriBound` (open)
H^1 enstrophy identity, `periodicEnstrophyIdentity`†	proved unconditionally
Small zero-mean H^1 data, `periodicTstar_eq_top_of_smallH1_of_rate`†	proved from named `PeriodicEnstrophyRateBound` + `PeriodicH1BlowupCriterion` (not formalized; [7, 8])
Ladyzhenskaya-type trilinear bound, `periodicH1TrilinearBound`†	proved (explicit unoptimized constant; Fourier/Agmon route; the inequality is classical: Ladyzhenskaya 1958 (2D), 1961 monograph (3D) [7])
Mean conservation, `periodicMeanConservation`†	proved unconditionally
Small zero-mean H^1 data, `periodicTstar_eq_top_of_smallH1_ladyzhenskaya`†	proved from `PeriodicH1BlowupCriterion` only (not formalized; [8])

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Earlier small-H^1 route, `integrability_of_smallH1`†	superseded: its hypothesis `PeriodicEnstrophyRiccati` ($\alpha > 0$) fails for constant flows (bug fixed in `ea5133d`)
Tether cubic rate, `clay_B_exact_of_tetherCubicRate`†	proved from `PeriodicTetherCubicRate` (open)
Uniform cubic rate vs. scaling, `uniform_cubic_rate_false_of_scaling`†	proved (abstract scaling step)
Direction route, `periodicBKMIntegrability_of_direction`†	proved from `PeriodicDirectionCriterion` (not formalized) + `PeriodicDirectionRegularity` (open)
Direction criterion, `periodicDirectionCriterion_of_growth`†	proved from `PeriodicDirectionEnstrophyGrowth` (CF 1993 [3] / BdV–B 2002 [2], open here) + `PeriodicH1BlowupCriterion` (Leray, open here)
Direction criterion (L^1 rate), `periodicDirectionCriterion_of_growthL1`†	proved from `PeriodicDirectionEnstrophyGrowthL1` (open here) + `PeriodicH1BlowupCriterion`; CF depletion factor `abs_cfDepletion_le` and Grönwall `gronwall_integrating_factor` proved
Bending-rate identity, `hasDerivAt_vortDir_lagrangian`†	proved: $D\mathbf{t}/Dt = (I - \mathbf{t}\otimes\mathbf{t})S\mathbf{t} + \nu\lvert\omega\rvert^{-1}(I - \mathbf{t}\otimes\mathbf{t})\Delta\omega$
Hidden property, `PeriodicBendingRateHiddenProperty`†	**open** (new formulation dated Oct 31 2026; precursors (geometric only, no regularity claims): the author's May 17/21 2026 notes (Π_u orthogonal to u, "invisible to the dynamics generated by H") and Aug 10 2026 notes *The Frohmanian Symplectic Tether* (B_{anti} via $\hat{u}\times(\cdot)$, B_{sym} with pointwise Π_u)); proved to imply `PeriodicDirectionRegularity`
Leray criterion, `periodicH1BlowupCriterion_of_smoothing`†	proved from `PeriodicH2SmoothingRate` + `PeriodicH3SmoothingRate` (open here) and the proved H^3 blow-up alternative
Energy equality, `periodicEnstrophyTimeIntegrable_of_cancellation`†	proved unconditionally: `periodicNonlinearEnergyCancellation` ($\langle\langle(u\cdot\nabla)u, u\rangle = 0$ [8, 10]) and hence `periodicEnstrophyTimeIntegrable`
Leray criterion, standard form, `periodicH1BlowupCriterion_of_smoothingStd`†	proved from `PeriodicH2SmoothingRateStd` + `PeriodicH2RootTimeIntegrable` + `PeriodicH3SmoothingRate` (open here [11, 9]); the pointwise `PeriodicH2SmoothingRate` is stronger than the standard estimate
L^1 growth, `periodicDirectionEnstrophyGrowthL1_of_holderStretching`†	proved from `PeriodicHolderStretchingEstimate` (open here) + `PeriodicEnstrophyTimeIntegrable` (proved)
$\kappa = 0$ tether, `tetherImpliesDirection_zero_iff`†	proved: headline Prop at $\kappa = 0 \Leftrightarrow$ `PeriodicDirectionRegularity`
Clay problem / NS global regularity	not proved, not claimed

A Theorem-to-Lean map

Names marked † were coined during formalization and are not the author's terminology; they may be renamed later.

Lean name	File	Repo @ commit

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periodicBKMHsControl_three†	PeriodicNSContinuation/BKMFinal.lean	pnscc @ ea5133d
periodicBKMCriterion†	PeriodicNSContinuation/BKMFinal.lean	pnscc @ ea5133d
clay_B_exact_of_integrability†	PeriodicNSContinuation/BKMFinal.lean	pnscc @ ea5133d
periodicBKMHsControl_three_of_lowFreq†	PeriodicNSContinuation/CommutatorAssembly.lean	pnscc @ ea5133d
periodicPhysicalCommutatorBound†	PeriodicNSContinuation/CommutatorAssembly.lean	pnscc @ ea5133d
periodicLowFreqVorticityBound†	PeriodicNSContinuation/LowFreqAssembly.lean	pnscc @ ea5133d
periodicBKMCriterion_of_hsControl	PeriodicNSContinuation/Continuation.lean	pnscc @ ea5133d
PeriodicBKMHsControl† (def)	PeriodicNSContinuation/Continuation.lean	pnscc @ ea5133d
clay_B_exact_of_bkm†	PeriodicNSContinuation/LocalExistenceProof.lean	pnscc @ ea5133d
periodicBKMIntegrability_iff_globalLifespan†	PeriodicNSContinuation/IntegrabilityAttack.lean	pnscc @ ea5133d
periodicEnstrophyIdentity†	PeriodicNSContinuation/EnstrophyIdentity.lean	pnscc @ ea5133d
periodicTstar_eq_top_of_smallH1_of_rate†	PeriodicNSContinuation/EnstrophyIdentity.lean	pnscc @ ea5133d
periodicH1TrilinearBound†	PeriodicNSContinuation/Ladyzhenskaya.lean	pnscc
periodicMeanConservation†	PeriodicNSContinuation/MeanConservation.lean	pnscc
clay_B_exact_of_tetherCubicRate†	PeriodicNSContinuation/TetherClaimAttack.lean	pnscc @ ea5133d
periodicBKMIntegrability_of_direction†	PeriodicNSContinuation/TetherClaimAttack.lean	pnscc @ ea5133d
tetherImpliesDirection_zero_iff†	PeriodicNSContinuation/TetherKappaZero.lean	pnscc @ ea5133d
Clay_B_exact (def)	FeffermanClayNS/FeffermanStatements.lean	dependency
PeriodicBKMCriterion, PeriodicBKMIntegrability (defs)	FeffermanClayNS/HighSetDepletionAndBKM.lean	dependency

pnscc = `BenFrohman/periodic-ns-classical-continuation` (public). "dependency" = `fefferman-clay-navier-stokes`, as pinned in the lake manifest.

B Axiom audit

At the current commit, `lake build` succeeds with 0 `sorry`, and `lake env lean AxiomAudit.lean` prints "AxiomAudit PASS: only propext, Classical.choice, Quot.sound". Every theorem above is in the audit, including those of `SmallDataAttack`, `TetherClaimAttack`, `EnstrophyIdentity` and `TetherKappaZero` (commits `052cef1`, `ea5133d`). The audit also covers `DirectionCriterion`, `BendingRate` and `ClassicalSteps`: the L^1 direction criterion `periodicDirectionCriterion_of_growthL1`, the bending-rate identity `hasDerivAt_vortDir_lagrangian`, the depletion bound `abs_cfDepletion_le`, the reductions `periodicDirectionRegularity_of_bending` (from the open bending-rate hypothesis `PeriodicBendingRateHiddenProperty`), `periodicH1BlowupCriterion_of_smoothing` and `periodicDirectionEnstrophyGrowthL1_of_holderStretching`; all use only `propext`,

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`Classical.choice`, `Quot.sound` (current commit; 0 `sorry`).

Counting convention. Two counts are reported, each with its method. (i) *Source count:* 755 declarations introduced by the keywords `theorem` or `lemma` (optionally preceded by attributes or modifiers) in the tracked `.lean` files of this repository, excluding `AxiomAudit.lean`. (ii) *Kernel count:* 867 non-internal theorem constants in the `PeriodicNSContinuation` module tree, as reported by `AxiomAudit.lean`; this includes theorems generated automatically by Lean. An earlier figure of 727 was not obtained by either method and is withdrawn; the figure 726 used in versions ≤ 8 was the source count before `EnergyCancellation.lean` was added.

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